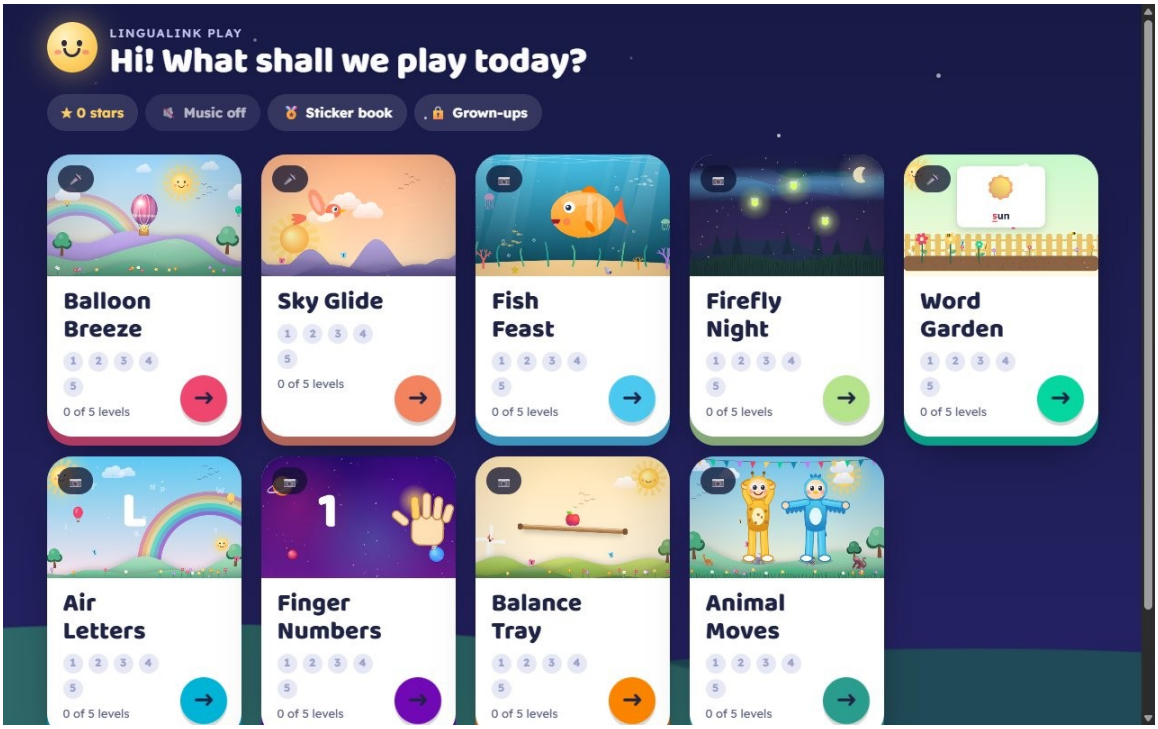
LinguaLink Play v5 Report

LINGUALINK360 THERAPY GAMES

Version 5 report: every feature, bug and fix from version 1 to version 5



4 November 2026 · 9 games · 45 levels · English and Urdu

For the project team, the clinical lead and the LinguaLink360 integration developers.

1. Summary

LinguaLink Play grew from five therapy games with a long list of problems into nine games with 45 levels, live animated art, voice and movement checking that runs in the browser, and progress a therapist can read level by level.

Status: version 5 is complete and all automated tests pass, so it is ready for a supervised pilot. It should not be called clinically proven or error free yet, because **no real child has used it**, and hand, finger, face and body tracking have only been tested with simulated positions (see section 9).

Version	Date	Theme	Games / levels	Main result
1.0.0	Before 31 Oct 2026	First version	5 / 25	Working games, but a QA review found 26 problems
2.0.0	31 Oct	Redesign for	5 / 25	All 26 QA problems fixed; splash, map,

Version	Date	Theme	Games / levels	Main result
	2026	children		stickers, Grown-ups area, offline use
3.0.0	1 Nov 2026	Hands-on feedback fixes	5 / 25	Calmer voice and pacing, music button, the game checks words itself, 7 hand-testing bugs fixed
4.0.0	4 Nov 2026	Therapy tuning and new games	9 / 45	Vowel recognition, level ladders, 4 new games, progress by level
5.0.0	4 Nov 2026	Look, writing, clean-up	9 / 45	Animated scenery, Buddy character, better air writing, two Word Garden modes, developer tools removed

Results at a glance

Measure	v1	v2	v3	v4	v5
Games	5	5	5	9	9
Levels	25	25	25	45	45
Camera game start	56 to 110 s	Progress bar, models bundled and reused	Same	Same, with a new body model	Same
Unit tests	Not recorded	30	30, plus Sky Glide steering tests	48	58
Browser checks	None	Added	Kept	39	40 of 40
Microphone recording checks	None	None	None	23	23
Word Garden checks	None	None	None	None	25 (3 with the real recogniser)

2. Version 1.0.0

The first version. Five games, 25 levels.

Features built

- **Balloon Breeze:** voice and breath support.
- **Sky Glide:** voice control, steered by pitch.

- **Fish Feast:** oral-motor moves with camera face tracking.
- **Firefly Night:** reaching and hand use with camera hand tracking.
- **Word Garden:** articulation practice with words, phrases and sentences.
- Sensors with a touch fallback for every game.
- Automatic difficulty inside a level.
- Results sent to LinguaLink360.

Problems found

A QA review of version 1 recorded 26 problems: 4 critical, 5 high, 9 medium and 8 low. They are listed with their fixes in the next section.

3. Version 2.0.0

31 October 2026 · commit f9e9760 · a redesign for children and a fix for every QA problem

Features built

- Splash screen with Lumi the guide and a loading bar, then "Tap to play".
- Child-friendly home: one big tile per game, stars earned, no level lists.
- A level map for each game. Levels open one by one, and locked levels say how to open them.
- Sticker book: 25 stickers, one per level, won with 2 or more stars.
- Rewards on the result screen: new sticker, "Level 2 is open!".
- Grown-ups area (hold 2 seconds): session reports with charts, Save as PDF, progress per game, choose any level, settings, clear data.
- A sound palette and gentle music for each game, never while a microphone game listens, plus varied praise.
- Balloon Breeze practises different sounds round by round.
- Sky Glide measures the voice range once per visit and reuses it.
- Installable app, offline play and a Node server for Render with security headers.
- A browser test suite and 11 new unit tests.

Bugs fixed

ID	Severity	Problem in v1	Fix
C1	Critical	Camera games took 56 to 110 s to start.	Camera library and models ship with the app, load with a progress bar, start loading when the map opens, and are reused for every level.
C2	Critical	Leaving during loading left a hidden game running that saved a fake result.	Closing now cancels loading and releases the camera.
C3	Critical	Word Garden scored only words answered: 2 of 5 gave 100 and 3 stars.	Scored against every item in the level.
C4	Critical	Pitch detection took 18 ms per frame.	About 0.4 ms, 20 times a

ID	Severity	Problem in v1	Fix
			second, in Sky Glide only.
H1	High	Word Garden could hear its own spoken word as the child.	The game's own speech is ignored.
H2	High	The result screen waited for the server.	Appears at once; unsent results are retried.
H3	High	Results were reported as sent when they could not be.	Delivery is now confirmed.
H4	High	Sky Glide reported an invented pitch range for a silent child.	Reports only what was measured.
H5	High	Urdu mode was mostly English.	Every string and game text translated.
M1	Medium	Balloon Breeze celebrated a failed round.	Celebrates only a success.
M2	Medium	Fish Feast therapist mode timed out after 6 to 8 s.	Waits, with Done and Not yet.
M3	Medium	Wrong letters highlighted for /s/ ("Wash", "is").	Only real target sounds are marked.
M4	Medium	Custom word lists ignored the level.	Lists can be per level or per position.
M5	Medium	Privacy note said "nothing is recorded" although the last try is kept.	Note corrected.
M6	Medium	All 25 levels listed at once with no progression.	Level map; levels open one by one.
M7	Medium	Could not be installed or used offline.	Installable app with offline play.
M8	Medium	Hub waited for Google Fonts before showing anything.	Fonts ship with the app.
M9	Medium	Phone hub was 4,193 px tall.	Redesigned home.
L1	Low	Countdown and calibration could not be paused.	They can.
L2	Low	"Auto difficulty" pushed a figure off the screen.	Layout fixed.
L3	Low	Firefly touch cursor stayed after the finger lifted.	It disappears.
L4	Low	Intro could name the wrong controls.	Names the controls in use.

ID	Severity	Problem in v1	Fix
L5	Low	Settings window was not keyboard friendly.	Native dialogs.
L6	Low	Reduced motion only affected menus.	Game scenery stands still too.
L7	Low	Unknown `?sound=` values were accepted.	Rejected.
L8	Low	Docs disagreed on the number of tests.	Corrected.

Other changes

- Results gained `result_id`, and `suggestion` can be `unfinished`.
- Start scripts use Node (`npm start`).

4. Version 3.0.0

1 November 2026 · commit 010a492 · fixes from playing the games by hand

Features built

- **Word Garden checks words itself** using the browser's speech recogniser, so play does not depend only on a parent. An adult can still override.
- **Music on/off button** on the home, map and game screens.
- Privacy defaults: online word check off and spoken instructions off unless a grown-up turns them on.
- Balloon Breeze has a "get ready" moment before each round listens.

Bugs fixed

Problem reported	Fix
The voice was irritating and kept interrupting.	Spoken instructions off by default; praise is shown, not spoken.
The level ring filled by itself.	Microphone logic reworked so background sound does not count as voice.
JSON notes showed on the screen.	Replaced with plain sentences.
The game moved too fast.	Longer rounds, celebrations and countdown.
The Sky Glide bird moved at random: "eee" up, "aaa" still up, sometimes no movement.	Steering rewritten and given its own tests. Replaced fully in v4.
New versions did not show in the browser.	Server checks each file on every load; the service worker fetches fresh code first, using the cache only offline.

Problem reported	Fix
Stars chimed during a hold and paused the microphone.	No chime while listening.

5. Version 4.0.0

2 November 2026 · commit 2e48925 · therapy tuning, four new games, visible progress

New games

Game	Therapy focus	How it works	Levels
Air Letters	Arm control, motor planning, letter shapes and sounds	Move an open hand along the letter. It pops up with a picture and is spoken: "B for Ball".	Straight lines, curves, zig zag, tricky letters, whole words (CAT, SUN, BUS)
Finger Numbers	Finger isolation, fine motor control, counting	A different finger each level writes numbers; then hold up 1 to 10 fingers.	Pointer, ring, thumb, little finger, show and count
Balance Tray	Two-handed coordination, load balancing, steadiness	Two hands hold a tray while fruit lands on it.	Level tray, two sides, heavy side, up and down, tower of treats
Animal Moves	Whole-body strength, balance, range of movement	Copy and hold giraffe, bird, palm tree, frog and flamingo poses, using body tracking.	Giraffe, palm, frog squats, flamingo balance, animal parade

These are original designs with similar therapy goals, not copies of another product. The middle finger is left out of Finger Numbers on purpose, because holding it up alone is a rude gesture in many places.

Features changed

Area	Before	After
Balloon Breeze	Any voice filled the balloon	Recognises the vowel. Only the sound asked for (aaa, eee, iii, ooo, uuu) blows it up, in visible steps. Noise, claps, hiss and whispers do nothing
Sky Glide	Steered by pitch	"eee" up, "aaa" down, silence straight. Quiet voices and whispers count. New way to play: Stop and say
Fish Feast	No clear progression	Holds grow from 1 to 6 s, levels ask for bigger movements, and pout detection also uses mouth narrowing
Word Garden	/s/, /k/, /p/	All sounds by default, adding /t/, /m/, /b/, /f/.

Area	Before	After
		Tracks results per sound
Progress	Scores only	Result card compares the main therapy number with last time. Grown-ups > Progress shows first and latest for each level
Home	One column on phones	Two columns on phones, smoother transitions
Tests	30 unit tests	48 unit tests, 39 browser checks, and a new test that plays recordings into a fake microphone

Bugs fixed

Problem	Fix
Saved settings kept Word Garden on /s/ only.	Settings migrated; a sound stays only if a grown-up picks it.
Short words such as "pig" were missed.	Shorter voice requirement, while a hiss alone still does not count.
A whisper flickered as a pitched voice and could fill the balloon.	A steady-voice check was added.
Starting a game could be overridden by the home screen.	Start-up race fixed.
"Hold up 1 fingers", "(1 sessions)", overlapping labels, poor contrast in Balance Tray, long phone home.	All fixed.
Difficulty moved the wrong way in tracing, balloon match and balance.	Direction corrected in each.

> ***Change for LinguaLink360:*** *Sky Glide no longer sends* `pitch_low_hz`, `pitch_high_hz` *or* `pitch_range_semitones`. *"mmm" was removed from Balloon Breeze because it is hard to tell from "uuu".*

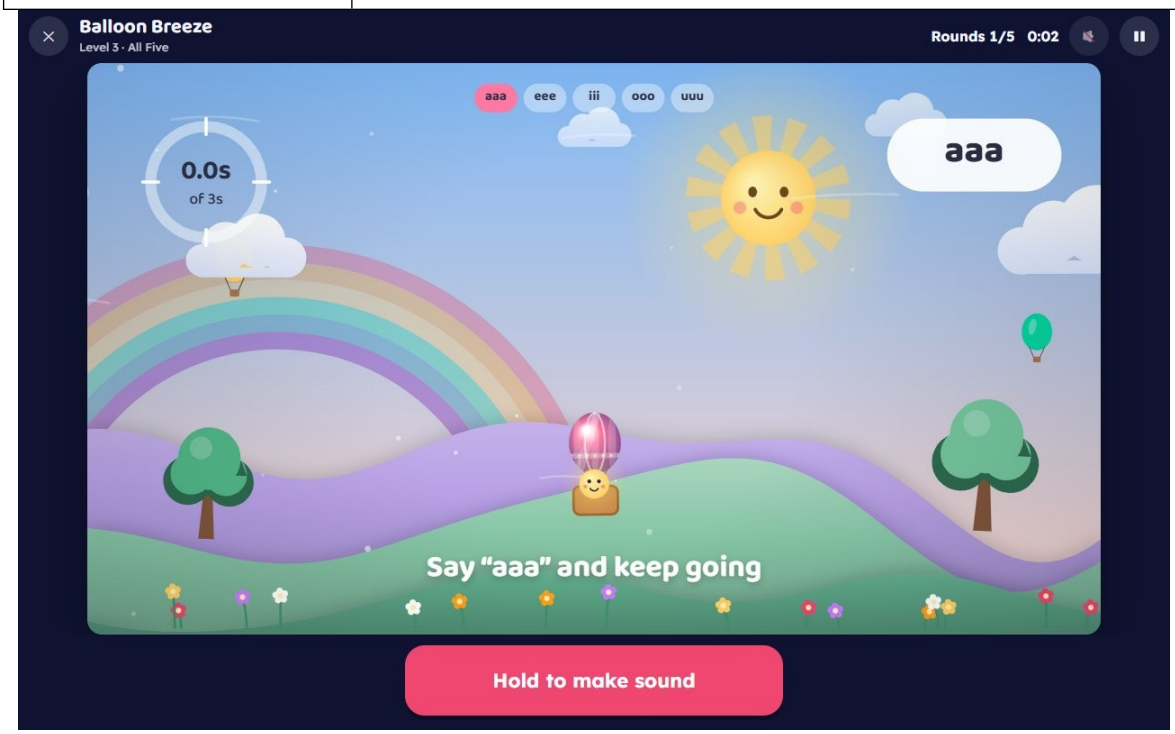
6. Version 5.0.0

4 November 2026 · not yet committed · look and feel, writing, two Word Garden modes, clean-up

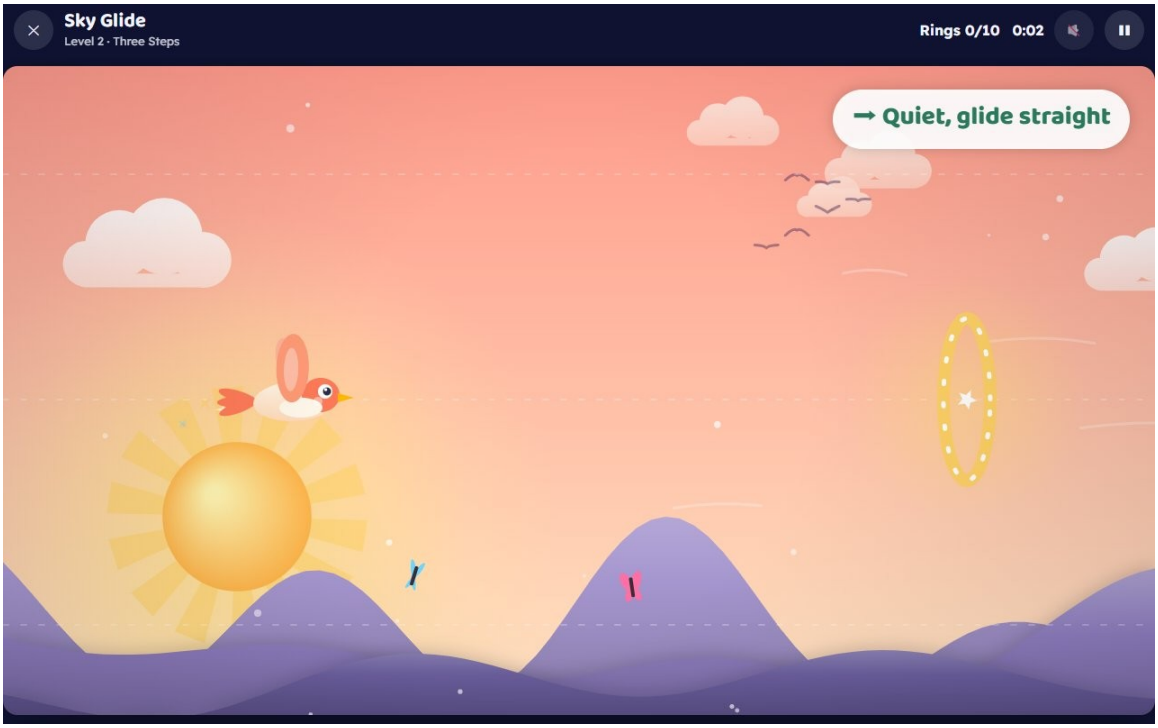
Animated graphics in every game

A shared animation kit keeps every scene moving. The home tiles are live too. "Less movement on screen" still stills the scenery.

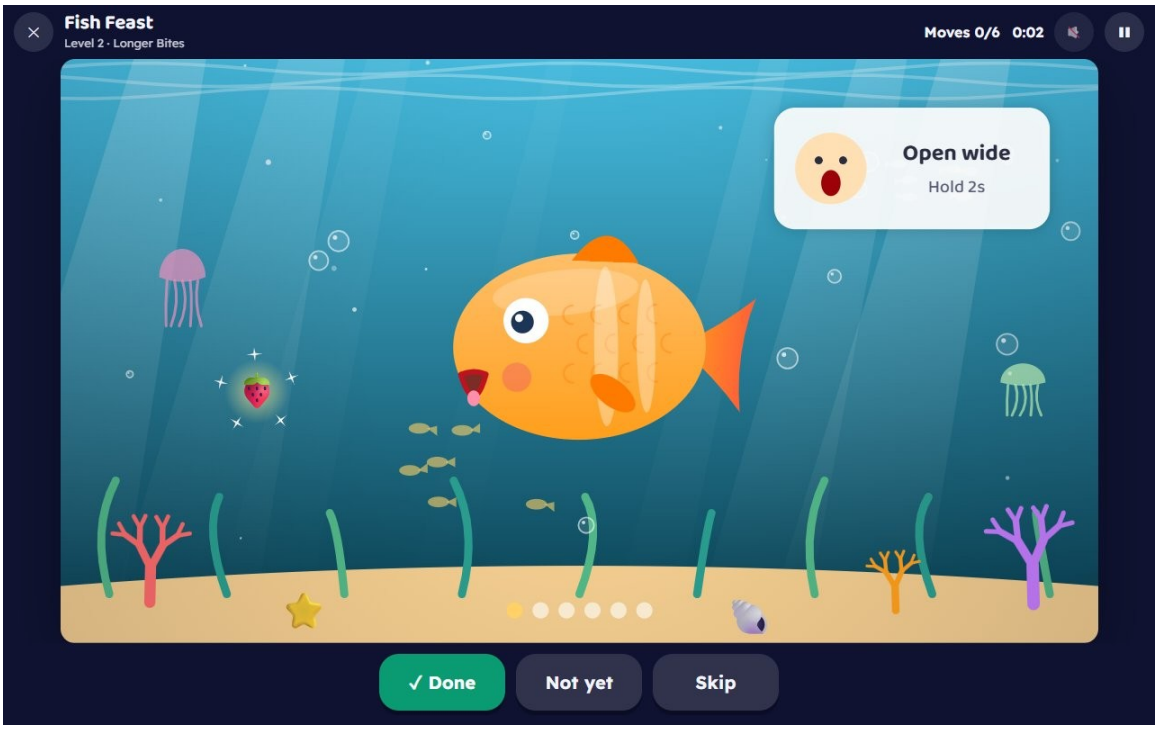
Game	What is new on screen
Balloon Breeze	Smiling sun, rainbow, far balloons, flocks, swaying trees, flowers, wind. Balloon has panels and a sliding shine, sparkles while the right sound is held, and a burst at each step
Sky Glide	Sunset with layered mountains and fast and slow clouds. A new flapping bird with a spark trail. Glowing rings with running lights
Fish Feast	Light shafts, shimmering surface, fish schools, jellyfish, coral, crab, bubbles. Pip blinks, paddles, swishes and has scales. The treat sparkles
Firefly Night	Northern lights, twinkling stars, shooting stars, crescent moon, trees, far fireflies, swaying grass, star cursor
Word Garden	Sun, clouds, butterflies, bees, flower beds. Each word card springs in and floats; rays and a ring of light when a flower blooms
Air Letters	Sky with rainbow, balloons and drifting letters. Magic slate, rainbow ink and a reward card with rays, confetti and dancing pictures
Finger Numbers	Galaxy with nebula, planets, rocket and shooting stars. The finger to use glows and wiggles; pictures pop in one by one as if counted
Balance Tray	Windmill, trees, flowers, butterflies, wooden tray, star markers on the hands
Animal Moves	Bunting, animal friends peeking over the hill, trees, flowers, and the new Buddy character



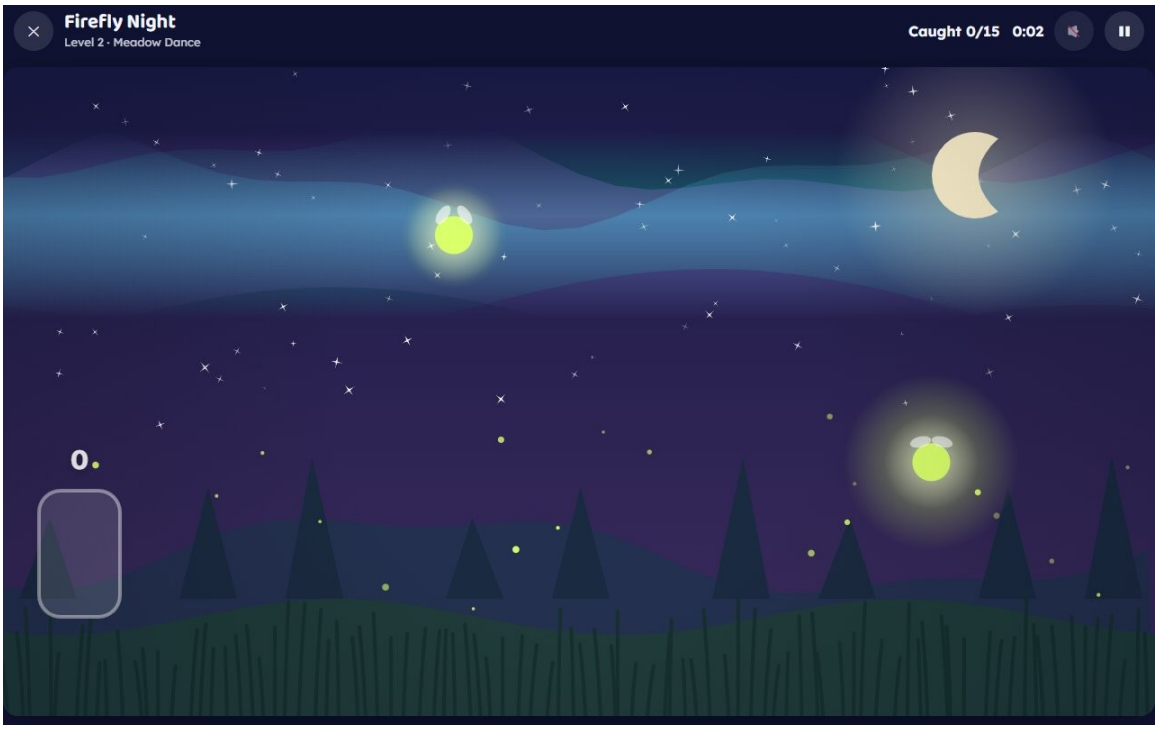
Balloon Breeze



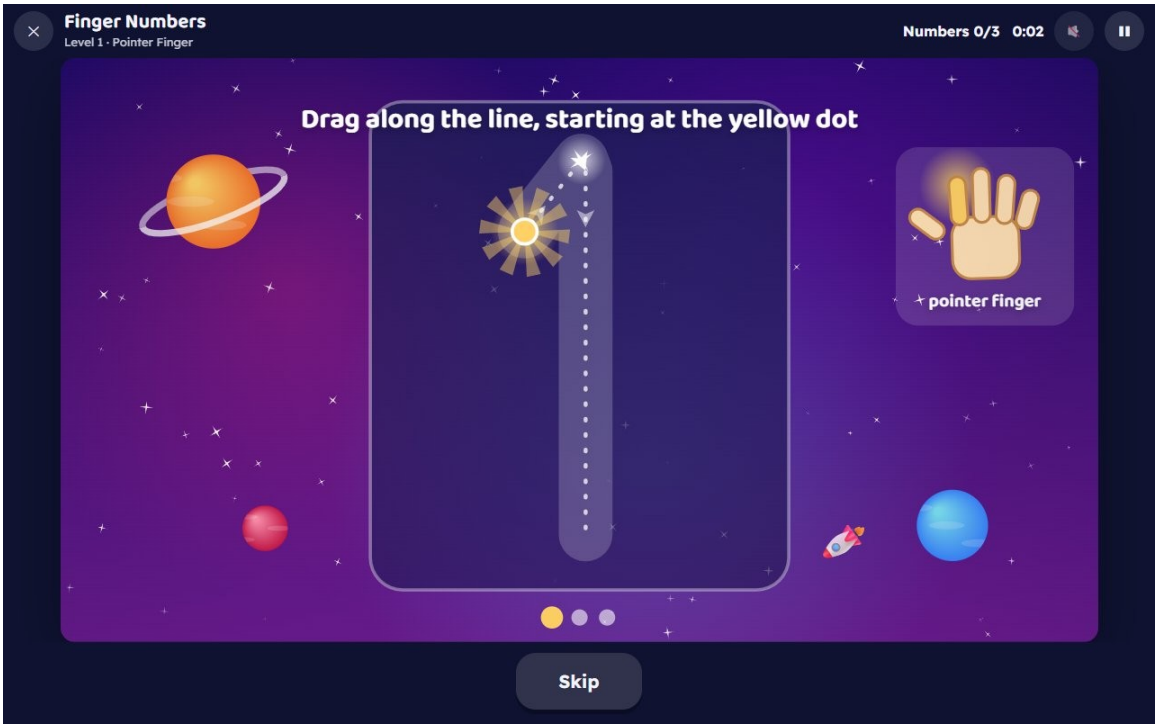
Sky Glide



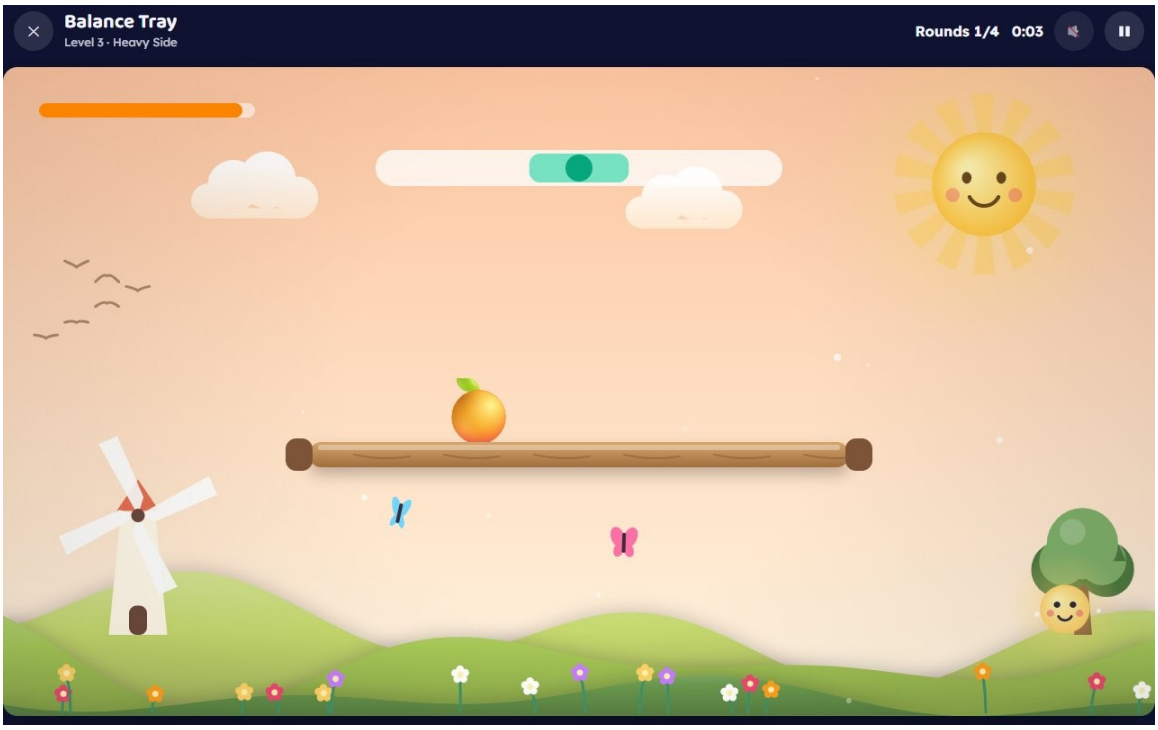
Fish Feast



Firefly Night



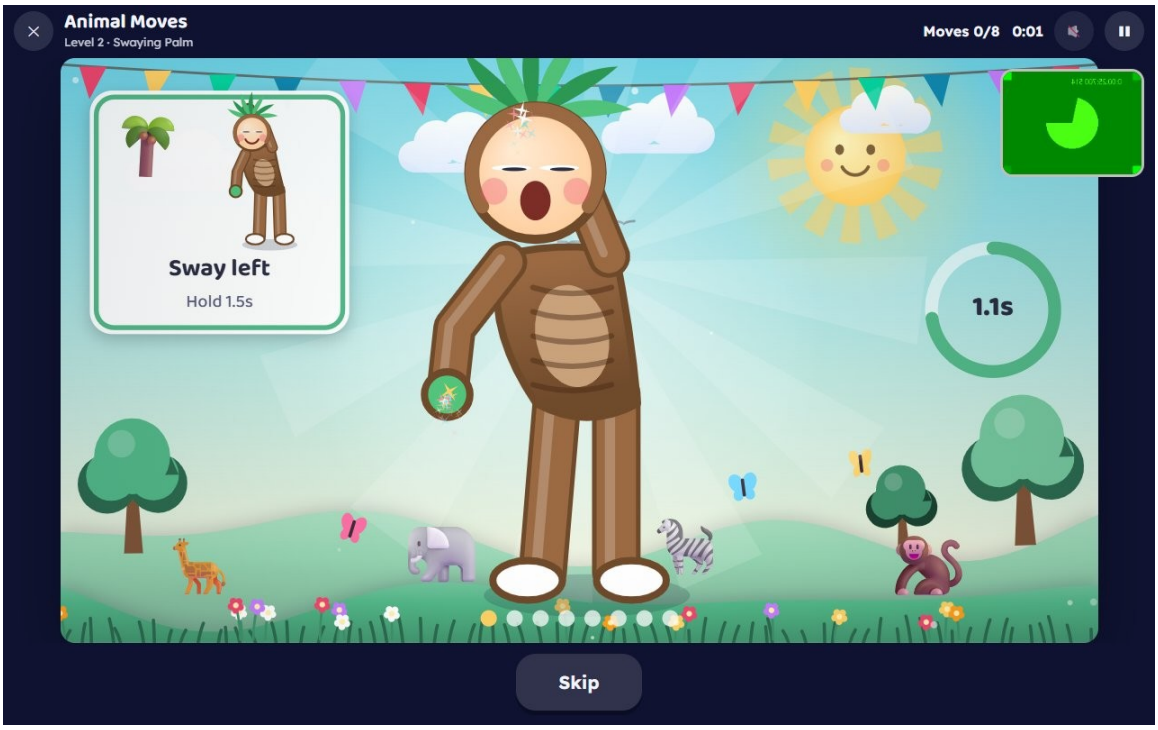
Finger Numbers



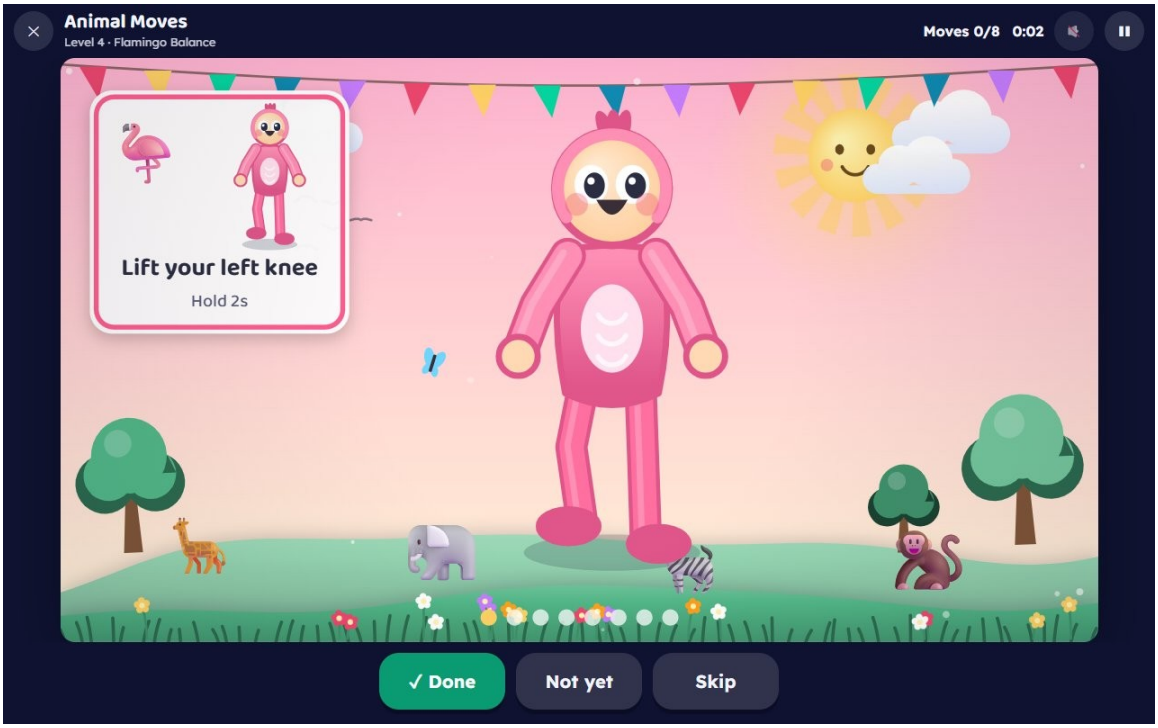
Balance Tray

Buddy replaces the stick figure

Animal Moves now shows Buddy, a round little kid in a giraffe, bird, palm tree, frog or flamingo costume. Buddy blinks and follows the child's body, glows and sends sparks from the hands during a pose, and shows each pose on the cue card. If the camera cannot see the feet, they are filled in so Buddy stays whole. In touch mode, Buddy demonstrates the pose.



Buddy following a simulated body in camera mode



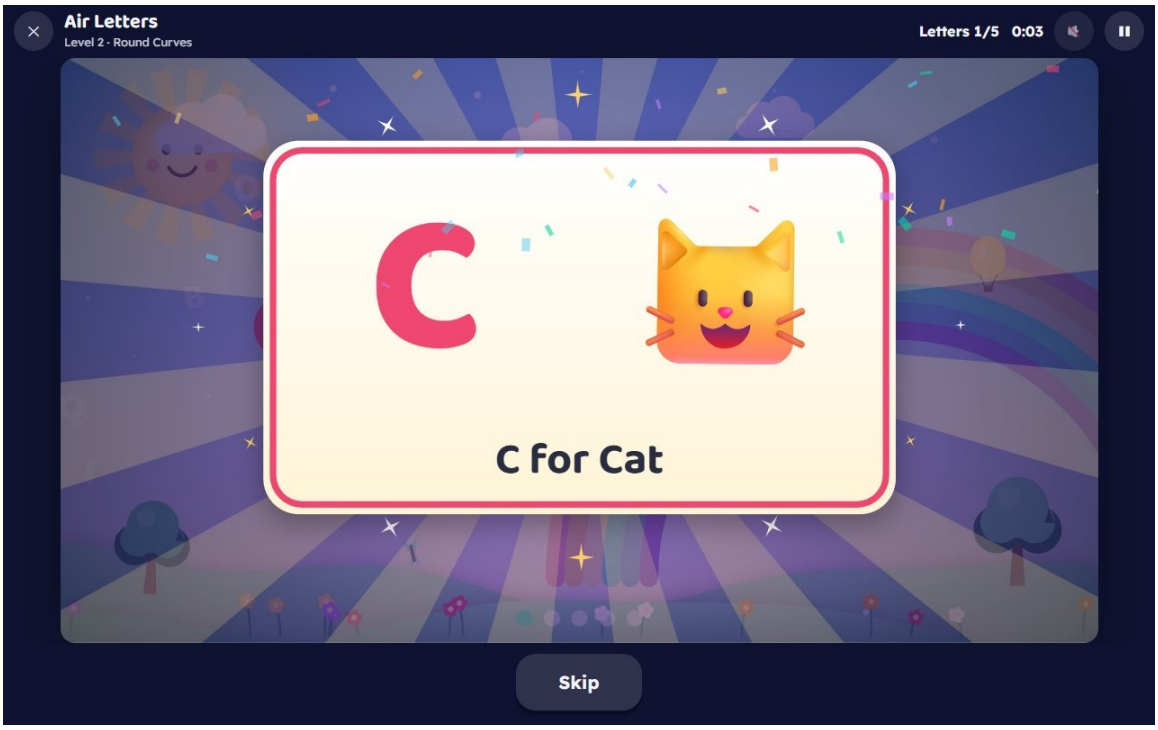
Flamingo costume showing the pose

Better writing with hand and finger

Problem	Fix
Camera shake made the line wobble	A filter smooths a slow hand and hardly touches a fast one, so there is no lag
A quick stroke was lost between camera frames	The game walks the path between frames, but a jump off the line earns little
Losing the hand for a moment lifted the pen	The pen stays down for 0.3 s
With two hands in view, the cursor could jump	The game stays on the hand that is writing
Sideways strokes ran faster than up and down	Movement is now equal in both directions
A finger pointing at the camera read as curled	Depth is now used
A stuck child had no help	After 9 s there is a little more room for error, up to twice, and a spark runs along the line
"Skip" did nothing on the reward card	It goes to the next item



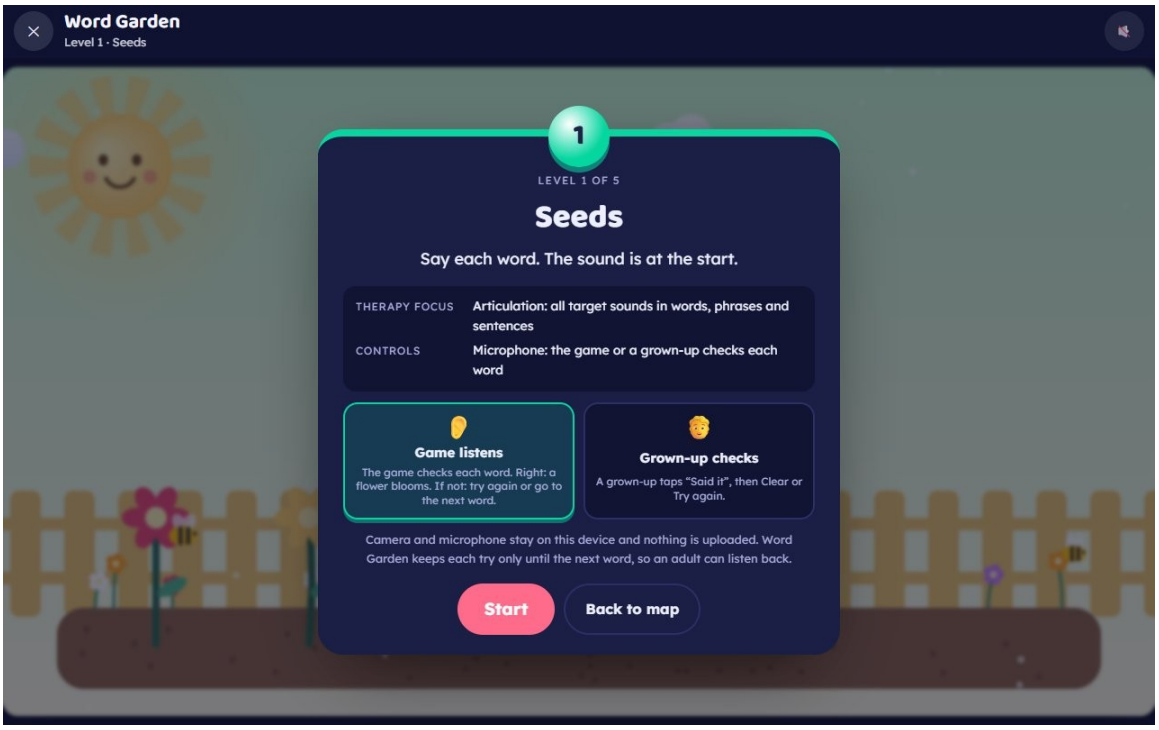
Rainbow ink, magic star, moving arrows



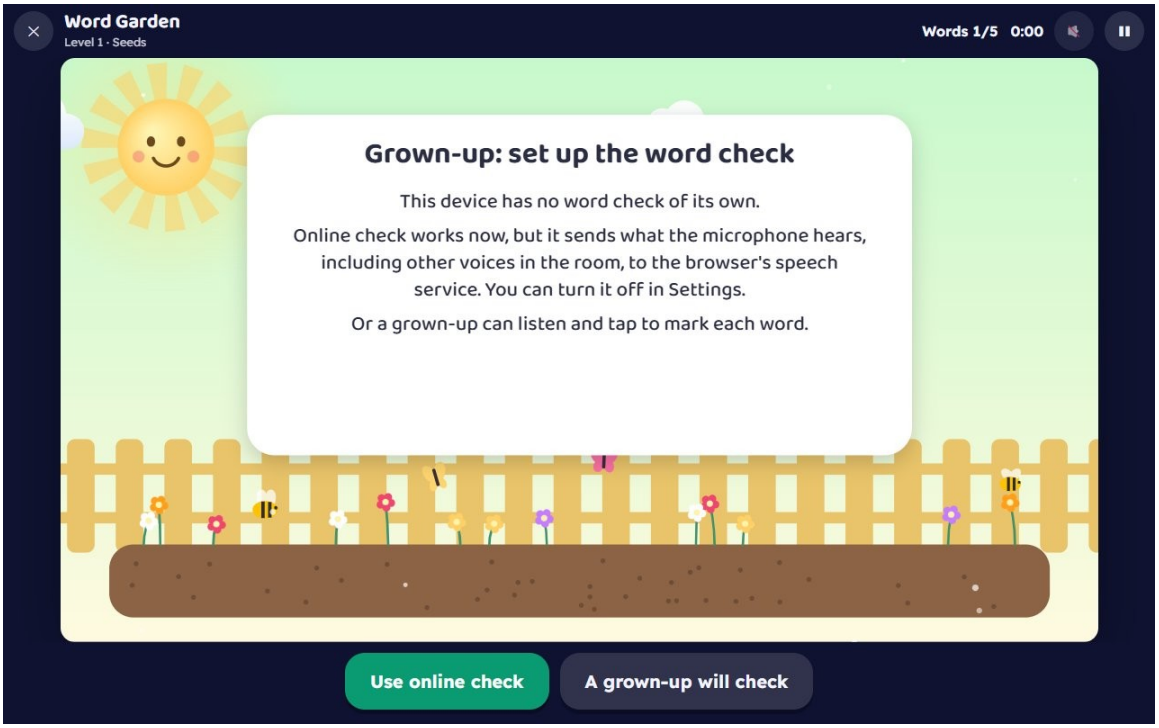
Reward card

Word Garden: two ways to play

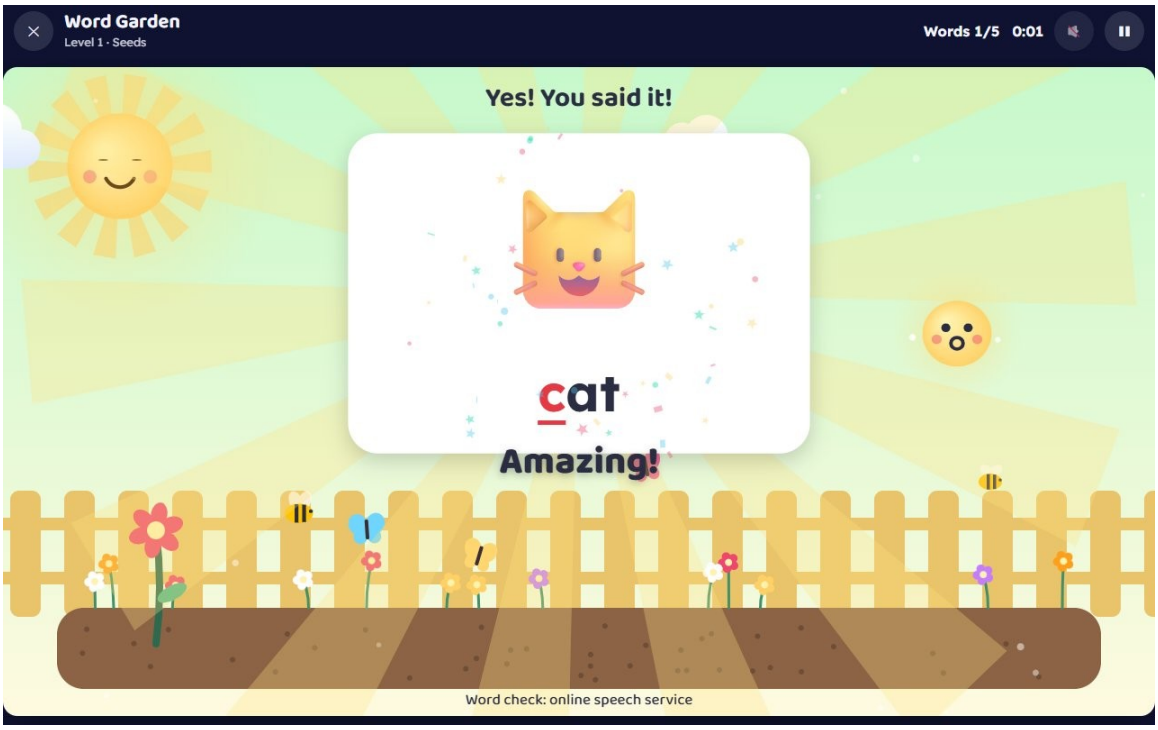
Way to play	How it works
Game listens	The game checks each word. Right: a flower blooms, then the next word comes. Wrong: "Not quite. Try again!" with what it heard, plus Hear it, Try again, Listen back, Correct and Next word. After 3 tries it moves on and keeps the word for practice
Grown-up checks	No recogniser runs. The adult taps Said it, then Clear or Try again. Nothing moves on until the adult taps. Touch mode always plays this way



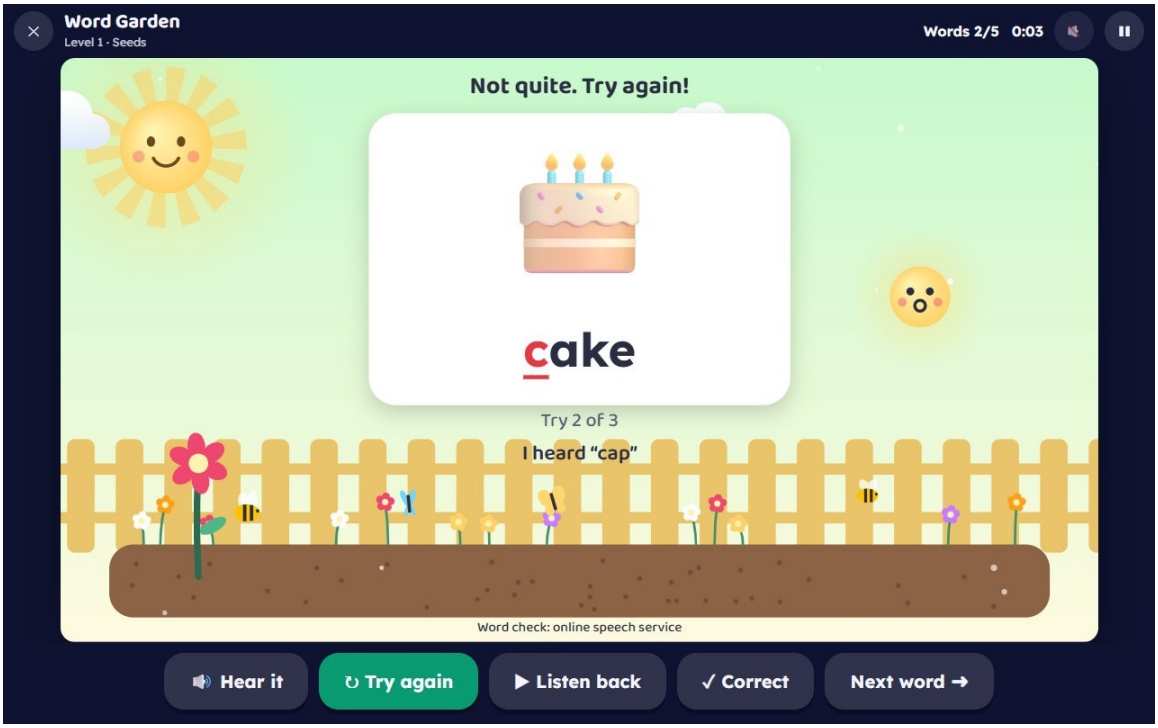
The choice on the level card



A grown-up decides before any audio goes online



Right word: a flower blooms



Wrong word: the prompt (simulated recogniser)

What testing the real recogniser showed

Test (recordings, Chrome 156)	Result
Online, sentence	Worked: "my cat is on the table", exact, 3 of 3. In the game, the sentence bloomed its flower, and the same recording was refused for the next sentence
Online, one short word ("cat")	Chrome heard speech but returned no words, 3 of 3
On-device pack, once installed	Returned no words in any test, and online then returned none in that browser profile either
Without a microphone track	The browser used the computer's real default microphone, not the game's. Fixed: the game now passes its own microphone

So the game no longer downloads the on-device pack. If the recogniser returns nothing before it has ever worked, the child is not marked down: a grown-up is asked instead, and failing that, the game moves on when the child speaks and says it could not check.

> ***Must be checked first:*** *single words with a real voice. The tests used synthetic speech, so it is unknown whether real children's single words return results. If they often return nothing, "Game listens" will not be dependable at levels 1 to 3, and a bundled offline recogniser should be evaluated.*

Developer tools removed

- JSON export buttons, on the session list and on each report.
- Technical settings names in session reports, and the result format and ID in the footer.
- Library and format names in About.
- Results still reach LinguaLink360 in the background, and Save as PDF remains.

Broken links and errors found and fixed

Problem	Fix
The public site served project files: `/docs`, `/integration`, Markdown files, `package.json`, the server code.	Blocked. Only the app is public. Takes effect on the next deploy.
The Urdu font was missing from the offline files.	Added.
The Laravel guide's samples accepted only 5 of the 9 games.	Corrected.
Word Garden counted one try too many after a miss.	Corrected.

Problem	Fix
A recogniser that hung could count against the child.	Replaced after 15 s, given up on after two hangs.
The camera preview covered the hold ring in Animal Moves.	Ring moved below it.
The word-check line was white on a pale background.	Now dark.
The privacy note said nothing is uploaded even with online check on.	A separate note now says so; it also warns that other voices in the room are sent.

A new automatic test now checks every linked file, element, offline file, script import and text string in both languages.

7. All nine games and their levels

Game	Level 1	Level 2	Level 3	Level 4	Level 5
Balloon Breeze	First Puffs: 2 s, aaa, eee	Three Sounds: 3 s	All Five: 3 s	Gentle Voice: 5 s, closer match	Over the Mountains: 7 s
Sky Glide	Two Clouds	Three Steps	Rolling Hills	Quick Swallow	Glide Slides
Fish Feast	Big Bites: 1 to 2 s	Longer Bites: 2 to 4 s	Happy Fish: open + smile	Bubble Pouts	Pufferfish Party: 3 to 6 s
Firefly Night	First Glow	Meadow Dance	Two-Hand Colours	Cross the Path	Number Trail
Word Garden	Seeds: sound at start	Sprouts: at end	Buds: in middle	Blooms: phrases	Full Garden: sentences
Air Letters	L T I H E	C O U J S	A V W N Z X	B D G P Q R K Y M F	CAT, SUN, BUS
Finger Numbers	Pointer: 1 2 3	Ring: 4 5 6	Thumb: 7 8 9	Little: 10 1 4 7	Show 1 to 10 fingers
Balance Tray	Level Tray	Two Sides	Heavy Side	Up and Down	Tower of Treats
Animal Moves	Tall Giraffe	Swaying Palm	Frog Squats	Flamingo Balance	Animal Parade

Features across versions

Feature	v1	v2	v3	v4	v5
Touch fallback, automatic difficulty	●	●	●	●	●
Results to LinguaLink360	●	● confirmed	●	●	●
Level map, stickers, Grown-ups area		●	●	●	●
Offline and installable		●	●	●	●
Full Urdu		●	●	●	●
Music button			●	●	●
Word Garden checks words itself			●	●	● with a choice
Vowel recognition				●	●
Progress by level, change since last time				●	●
Hand writing, finger numbers, balance, body games				●	● improved
Animated scenery and live home tiles					●
Buddy character					●
JSON export		●	●	●	Removed

8. Test evidence for version 5

Suite	Result	What it covers
Unit tests	58 of 58	Vowels, tracing, poses, game rules, scoring, reports, links, text in both languages
Browser checks	40 of 40	All 45 levels start, play and finish; camera modes, offline, Urdu, phone width, v1 bugs
Microphone recordings	23 of 23	Generated voices through a fake microphone: vowels, whispers, claps, full games
Word Garden	25 of 25	Both ways to play with a stand-in recogniser, plus 3 checks with the real online recogniser using a synthetic recording

On a test machine with no graphics acceleration, games ran at about 33 to 49 frames per second in most runs, with one measured dip to 25 in Balloon Breeze, and the home screen at 29. A device with a graphics chip should do better, but this has not been measured on tablets or phones.

9. What still needs work

Not tested yet

Not tested	Why it matters	What to do
Real children	Therapy value, enjoyment, fatigue and thresholds are all unproven	Supervised pilot with a clinician
Real voices with Word Garden, especially single words	Synthetic single words returned nothing from Chrome	Try levels 1 to 3 with real voices
Real voices with vowels	Tested on generated voices. eee/iii and ooo/uuu sound alike	Calibrate with children
Real people on camera	Hand, finger, face and body tracking, and the new writing filter, were tested only with simulated positions	Run the checklist in TESTING.md
Phones and tablets	Speech and frame rate may differ. Android may not share a microphone between the game and the recogniser	Test on the target devices
Updating an installed older version	Service worker upgrade not tested end to end	Open the live site after deploying

Decisions to confirm

Decision	Current setting	Who decides
Online word check sends what the microphone hears, including other voices, to the browser's speech service (Google in Chrome). This needs review for children's data.	Off until a grown-up allows it. A LinguaLink360 page can forbid it	Clinical lead and privacy owner
If single words do not work, evaluate a bundled offline recogniser. It would keep audio on the device, but adds a large download and is unproven here.	Not built	Project team
JSON export is gone. Data now leaves only through LinguaLink360 or Save as PDF.	Removed as requested	Therapists using the games

Decision	Current setting	Who decides
		standalone
Level targets, hold times and word lists	Set from general practice, not clinically reviewed	Clinical lead
Urdu text, including the new v5 text	Needs native-speaker review	Project team

Before the live site changes

Version 5 is not committed or deployed yet. The live site keeps running the last pushed version until these changes are pushed.